LAB SESSION 05

PROGRAM FLOW CONTROL USING JUMP

1. **Write a program to display the first letter of your name 10 times using jump instruction. For example, your name starts with "A" then the output will be:**

AAAAAA

Source Code:

```
TITLE LAB 5 ACTIVITY 1       ;Title of the program
.MODEL SMALL                 ;Define memory model
.STACK 100H                  ;Set the size of the stack
.CODE                        ;Code segment Starts

MAIN PROC
    MOV CX, 10               ;Initialize CL With The Count (10)
    MOV AL, 'S'              ;Initialize AL With The Character 'S'
LOOP1:
    MOV AH,2                 ;Set AH to 2, indicating the "print character" function
    MOV DL,AL                ;Move the character from AL to DL
    INT 21H                  ;Call the DOS interrupt to print the character

    DEC CX                   ;Decrement the loop counter

    JNZ LOOP1                ;Jump back to LOOP1 if CX is not zero

    MOV AH, 4CH              ;Set AH to 4CH, indicating the "terminate program" function
    INT 21H                  ;Call the DOS interrupt to exit to DOS

MAIN ENDP                    ;End Of The Main Procedure
END MAIN                     ;End Of The Entire Program
```

OUTPUT:

2. **Write a program to display all the ASCII characters on screen line by line. Make sure to display character screen-by-screen.**

   (Hint: After displaying first 23 characters, ask user to continue by providing the statement "press any key to continue..." to display next 23 characters and so on.)

Source Code:

```
TITLE LAB 5 ACTIVITY 2      ;Title of the program
.MODEL SMALL                ;Define memory model
.STACK 100H                 ;Set the size of the stack
.DATA                       ;Declare a section of memory for storing data

PROMPT DB 'THE 256 ASCII CHARACTER ARE: $'
CONTINUE DB 'PRESS ANY KEY TO CONTINUE... $'

.CODE                       ;Code Segment starts
MAIN PROC

    ; INITIATE DATA SEGMENT
    MOV AX, @DATA
    MOV DS, AX

    ; DISPLAY PROMPT
    MOV AH, 9
    MOV DX, OFFSET PROMPT
    INT 21H

    ; NEXT LINE
    MOV AH, 2
    MOV DL, 0AH
    INT 21H

    ; MOVE THE CURSOR TO THE BEGINNING OF THE CURRENT LINE
    MOV DL, 0DH
    INT 21H

    MOV CX, 23
    ; Initialize DL to 0
    MOV DL, 0

LOOP1:
     MOV AH, 2
    INT 21H
    ;Increment DL
    INC DL
    ;Decrement CX
    DEC CX
    JNZ LOOP1

    MOV AH, 2
    MOV DL, 0AH
    INT 21H

    MOV DL, 0DH
    INT 21H

    ; Display continue message
    MOV AH, 9
    MOV DX, OFFSET CONTINUE
    INT 21H

    ; Wait for user input
    MOV AH, 1
    INT 21H
```

```

    ; Clear screen
    MOV AH, 6
    MOV AL, 0
    MOV BH, 7
    MOV CX, 0
    MOV DX, 184FH
    INT 10H

    MOV CX, 23
    MOV DL, 18H

LOOP2:
      MOV AH, 2
    INT 21H

    INC DL

    DEC CX
    JNZ LOOP2

    MOV AH, 2
    MOV DL, 0AH
    INT 21H

    MOV DL, 0DH
    INT 21H

    ; Display continue message
    MOV AH, 9
    MOV DX, OFFSET CONTINUE
    INT 21H

    ; Wait for user input
    MOV AH, 1
    INT 21H

    ; Clear screen
    MOV AH, 6
    MOV AL, 0
    MOV BH, 7
    MOV CX, 0
    MOV DX, 184FH
    INT 10H

    MOV CX, 23
    MOV DL, 2FH

LOOP3:
    MOV AH, 2
    INT 21H
    INC DL
    DEC CX
    JNZ LOOP3

```

```
    MOV AH, 2
    MOV DL, 0AH
    INT 21H
    MOV DL, 0DH
    INT 21H

    ; DISPLAY CONTINUE MESSAGE
    MOV AH, 9
    MOV DX, OFFSET CONTINUE
    INT 21H

    ; WAIT FOR USER INPUT
    MOV AH, 1
    INT 21H

    ; CLEAR SCREEN
    MOV AH, 6
    MOV AL, 0
    MOV BH, 7
    MOV CX, 0
    MOV DX, 184FH
    INT 10H

    MOV CX, 23
    MOV DL, 46H

LOOP4:
    MOV AH, 2
    INT 21H
    INC DL
    DEC CX
    JNZ LOOP4

    MOV AH, 2
    MOV DL, 0AH
    INT 21H
    MOV DL, 0DH
    INT 21H

    ; DISPLAY CONTINUE MESSAGE
    MOV AH, 9
    MOV DX, OFFSET CONTINUE
    INT 21H

    ; WAIT FOR USER INPUT
    MOV AH, 1
    INT 21H

    ; CLEAR SCREEN
    MOV AH, 6
    MOV AL, 0
    MOV BH, 7
    MOV CX, 0
    MOV DX, 184FH
    INT 10H
```

```

    MOV CX, 23
    MOV DL, 5DH

LOOP5:
    MOV AH, 2
    INT 21H
    INC DL
    DEC CX
    JNZ LOOP5

    MOV AH, 2
    MOV DL, 0AH
    INT 21H
    MOV DL, 0DH
    INT 21H

    ; DISPLAY CONTINUE MESSAGE
    MOV AH, 9
    MOV DX, OFFSET CONTINUE
    INT 21H

    ; WAIT FOR USER INPUT
    MOV AH, 1
    INT 21H

    ; CLEAR SCREEN
    MOV AH, 6
    MOV AL, 0
    MOV BH, 7
    MOV CX, 0
    MOV DX, 184FH
    INT 10H

    MOV CX, 23
    MOV DL, 74H

LOOP6:
    MOV AH, 2
    INT 21H
    INC DL
    DEC CX
    JNZ LOOP6

    MOV AH, 2
    MOV DL, 0AH
    INT 21H
    MOV DL, 0DH
    INT 21H

    ; DISPLAY CONTINUE MESSAGE
    MOV AH, 9
    MOV DX, OFFSET CONTINUE
    INT 21H

    ; WAIT FOR USER INPUT
    MOV AH, 1
```

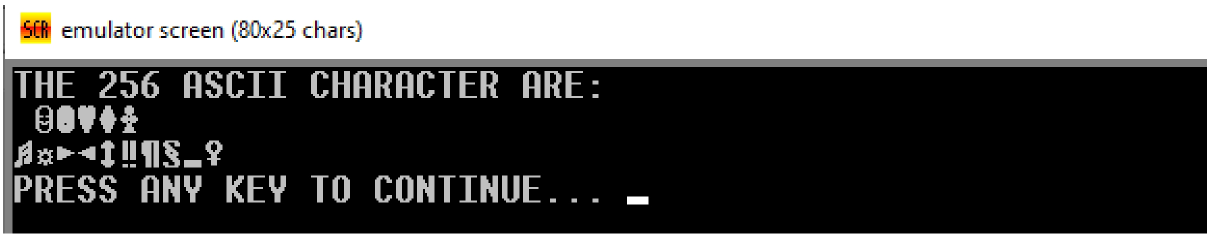
```
    INT 21H

    ; CLEAR SCREEN
    MOV AH, 6
    MOV AL, 0
    MOV BH, 7
    MOV CX, 0
    MOV DX, 184FH
    INT 10H

    MOV CX, 23
    MOV DL, 8BH

LOOP7:
    MOV AH, 2
    INT 21H
    INC DL
    DEC CX
    JNZ LOOP7

    MOV AH, 2
    MOV DL, 0AH
    INT 21H
    MOV DL, 0DH
    INT 21H

    ; DISPLAY CONTINUE MESSAGE
    MOV AH, 9
    MOV DX, OFFSET CONTINUE
    INT 21H

    ; WAIT FOR USER INPUT
    MOV AH, 1
    INT 21H

    ; CLEAR SCREEN
    MOV AH, 6
    MOV AL, 0
    MOV BH, 7
    MOV CX, 0
    MOV DX, 184FH
    INT 10H

    MOV CX, 23
    MOV DL, 0A2H

LOOP8:
    MOV AH, 2
    INT 21H
    INC DL
    DEC CX
    JNZ LOOP8

    MOV AH, 2
    MOV DL, 0AH
    INT 21H
    MOV DL, 0DH
```

```
    INT 21H

    ; DISPLAY CONTINUE MESSAGE
    MOV AH, 9
    MOV DX, OFFSET CONTINUE
    INT 21H

    ; WAIT FOR USER INPUT
    MOV AH, 1
    INT 21H

    ; CLEAR SCREEN
    MOV AH, 6
    MOV AL, 0
    MOV BH, 7
    MOV CX, 0
    MOV DX, 184FH
    INT 10H

    MOV CX, 23
    MOV DL, 0B8H

LOOP9:
    MOV AH, 2
    INT 21H
    INC DL
    DEC CX
    JNZ LOOP9

    MOV AH, 2
    MOV DL, 0AH
    INT 21H
    MOV DL, 0DH
    INT 21H

    ; DISPLAY CONTINUE MESSAGE
    MOV AH, 9
    MOV DX, OFFSET CONTINUE
    INT 21H

    ; WAIT FOR USER INPUT
    MOV AH, 1
    INT 21H

    ; CLEAR SCREEN
    MOV AH, 6
    MOV AL, 0
    MOV BH, 7
    MOV CX, 0
    MOV DX, 184FH
    INT 10H

    MOV CX, 23
    MOV DL, 0B9H

```

```
LOOP10:
      MOV AH, 2
      INT 21H
      INC DL
      DEC CX
      JNZ LOOP10

      MOV AH, 2
      MOV DL, 0AH
      INT 21H
      MOV DL, 0DH
      INT 21H

      ; DISPLAY CONTINUE MESSAGE
      MOV AH, 9
      MOV DX, OFFSET CONTINUE
      INT 21H

      ; WAIT FOR USER INPUT
      MOV AH, 1
      INT 21H

      ; CLEAR SCREEN
      MOV AH, 6
      MOV AL, 0
      MOV BH, 7
      MOV CX, 0
      MOV DX, 184FH
      INT 10H

      MOV CX, 23
      MOV DL, 0D0H

LOOP11:
      MOV AH, 2
      INT 21H
      INC DL
      DEC CX
      JNZ LOOP11

      MOV AH, 2
      MOV DL, 0AH
      INT 21H
      MOV DL, 0DH
      INT 21H

      ; DISPLAY CONTINUE MESSAGE
      MOV AH, 9
      MOV DX, OFFSET CONTINUE
      INT 21H

      ; WAIT FOR USER INPUT
      MOV AH, 1
      INT 21H

      ; CLEAR SCREEN
      MOV AH, 6
```

```
    MOV AL, 0
    MOV BH, 7
    MOV CX, 0
    MOV DX, 184FH
    INT 10H

    MOV CX, 23
    MOV DL, 0E7H

LOOP12:
    MOV AH, 2
    INT 21H
    INC DL
    DEC CX
    JNZ LOOP12

    MOV AH, 2
    MOV DL, 0AH
    INT 21H
    MOV DL, 0DH
    INT 21H

    ; DISPLAY CONTINUE MESSAGE
    MOV AH, 9
    MOV DX, OFFSET CONTINUE
    INT 21H

    ; WAIT FOR USER INPUT
    MOV AH, 1
    INT 21H
    ; CLEAR SCREEN
    MOV AH, 6
    MOV AL, 0
    MOV BH, 7
    MOV CX, 0
    MOV DX, 184FH
    INT 10H
    MOV CX, 3
    MOV DL, 0FEH
LOOP13:
    MOV AH, 2
    INT 21H
    INC DL
    DEC CX
    JNZ LOOP13
    ;EXIT TO DOS
    MOV AH,4CH
    INT 21H

MAIN ENDP                       ;End Of The Main Procedure
END MAIN                        ;End Of The Entire Program
```

OUTPUT:



```
↑↓→←∟↔▲   !"#$%&'()*+,-.
PRESS ANY KEY TO CONTINUE... _
```

```
/0123456789:;<=>?@ABCDE
PRESS ANY KEY TO CONTINUE...
```

```
FGHIJKLMNOPQRSTUVWXYZ[\
PRESS ANY KEY TO CONTINUE...
```

```
]^_`abcdefghijklmnopqrs
PRESS ANY KEY TO CONTINUE...
```

```
tuvwxyz{¦}~⌂Çüéâäàåçêëè
PRESS ANY KEY TO CONTINUE...
```

```
ïîìÄÅÉæÆôöòûùÿÖÜ¢£¥₧ƒáí
PRESS ANY KEY TO CONTINUE...
```

```
óúñÑªº¿⌐¬½¼¡«»░▒▓│┤╡╢╖╕
PRESS ANY KEY TO CONTINUE...
```

```
╕╣║╗╝╜╛┐└┴┬├─┼╞╟╚╔╩╦╠═╬
PRESS ANY KEY TO CONTINUE... _
```

```
╣║╗╝╜╛┐└┴┬├─┼╞╟╚╔╩╦╠═╬╧
PRESS ANY KEY TO CONTINUE...
```

```
╨╤╥╙╘╒╓╫╪┘┌█▄▌▐▀αßΓπΣσµ
PRESS ANY KEY TO CONTINUE... _
```

```
τΦΘΩδ∞φε∩≡±≥≤⌠⌡÷≈°∙·√ⁿ²
PRESS ANY KEY TO CONTINUE...
```

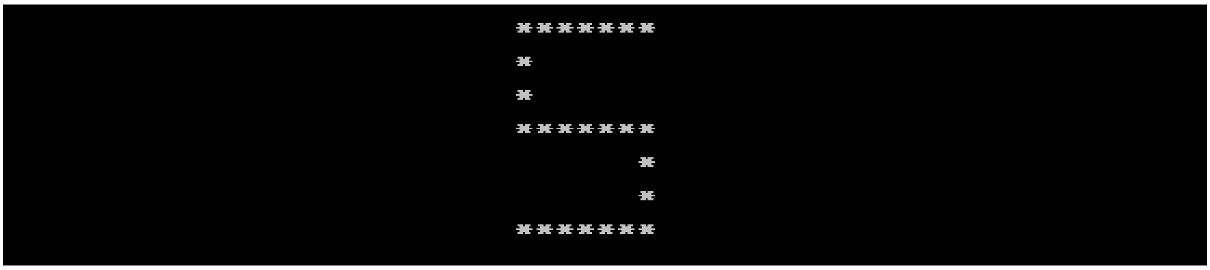
3. **Write a program to display the first letter of your name, using the character "*", at the center of screen. For example, if the first letter of your name is S, then following pattern should be displayed.**

Source Code:

```
TITLE LAB 5 ACTIVITY 3      ;Title of the program
.MODEL SMALL                ;Define memory model
.STACK 100H                 ;Set the size of the stack
.CODE                       ;Code segment Starts

MAIN PROC                   ;Start Main Procedure
    MOV AX, 2               ;To Set Video Mode 2 (Monochrome)
    INT 10H

    MOV DX, 0727H           ;To Set display page and Cursor Position
    MOV AH, 2
    INT 10H

    MOV CX, 4               ;To Initialize Loop Counter

LOOP1:
    MOV AH, 2               ;To Display '*'
    MOV DL, '*'
    INT 21H

    MOV AH, 2               ;Move To The Next Line
    MOV DL, 0AH
    INT 21H


    MOV AH, 2               ;Move Cursor To The Beginning Of The Line
    MOV DL, 08H
    INT 21H

    DEC CX                  ;To Decrement Loop Counter


    JNZ LOOP1               ;Repeat Loop1 Until CX Become Zero


    MOV DX, 0727H           ;To Set display page and Cursor Position Again
    MOV AH, 2
    INT 10H

    MOV CX, 7               ;To Set A New Loop Counter

LOOP2:
    MOV AH, 2               ;To Display '*'
    MOV DL, '*'
    INT 21H

    DEC CX                  ;To Decrement Loop Counter

    JNZ LOOP2               ;Repeat Loop2 Until CX Become Zero

    MOV DX, 0B27H           ;To Set display page and Cursor Position Again
    MOV AH, 2
    INT 10H

```

```
    MOV CX, 7              ;To Set A New Loop Counter

    MOV DX, 0A2DH          ;To Set display page and Cursor Position Again
    MOV AH, 2
    INT 10H

    MOV CX, 4              ;To Set A New Loop Counter

LOOP3:
    MOV AH, 2              ;To Display '*'
    MOV DL, '*'
    INT 21H

    MOV AH, 2              ;Move To The Next Line
    MOV DL, 0AH
    INT 21H

    MOV AH, 2              ;Move Cursor To The Beginning Of The Line
    MOV DL, 08H
    INT 21H

    DEC CX                 ;To Decrement Loop Counter

    JNZ LOOP3              ;Repeat Loop3 Until CX Become Zero

    MOV DX, 0A27H          ;To Set display page and Cursor Position Again
    MOV AH, 2
    INT 10H

    MOV CX, 7              ;To Set A New Loop Counter

LOOP4:
    MOV AH, 2              ;To Display '*'
    MOV DL, '*'
    INT 21H

    MOV AH, 2              ;Move To The Next Line
    MOV DL, 0H
    INT 21H

    MOV AH, 2              ;Move Cursor To The Beginning Of The Line
    MOV DL, 08H
    INT 21H

    DEC CX                 ;To Decrement Loop Counter

    JNZ LOOP4              ;Repeat Loop4 Until CX Become Zero


    MOV DX, 0D27H          ;To Set display page and Cursor Position Again
    MOV AH, 2
    INT 10H

    MOV CX, 7              ;To Set A New Loop Counter

LOOP5:
    MOV AH, 2              ;To Display '*'
    MOV DL, '*'
    INT 21H

    DEC CX                 ;To Decrement Loop Counter

    JNZ LOOP5              ;Repeat Loop4 Until CX Become Zero

    MOV AH, 4CH            ;Exit To DOSBOX
    INT 21H

MAIN ENDP                  ;End Of The Main Procedure
END MAIN                   ;End Of The Entire Program
```

OUTPUT:



4. Fill out the missing value, using target address calculation, in the following table.

IP	Opcode	Code
E06C	733F	JNZ CODE1
E06E	……….	MOV AX,3456H
……….	……….	………………….
E072	7139	JNO CODE1
E074	………..	MOV CX, 100H
……….	…………	…………………….
E08C	8ED8	C8: MOV DS, AX
………..	………..	…………………..
EOA7	EBE3	JMP C8
E0A9	………..	CMP AX
………..	…………	………………….
E0AD	F4	CODE1: HLT